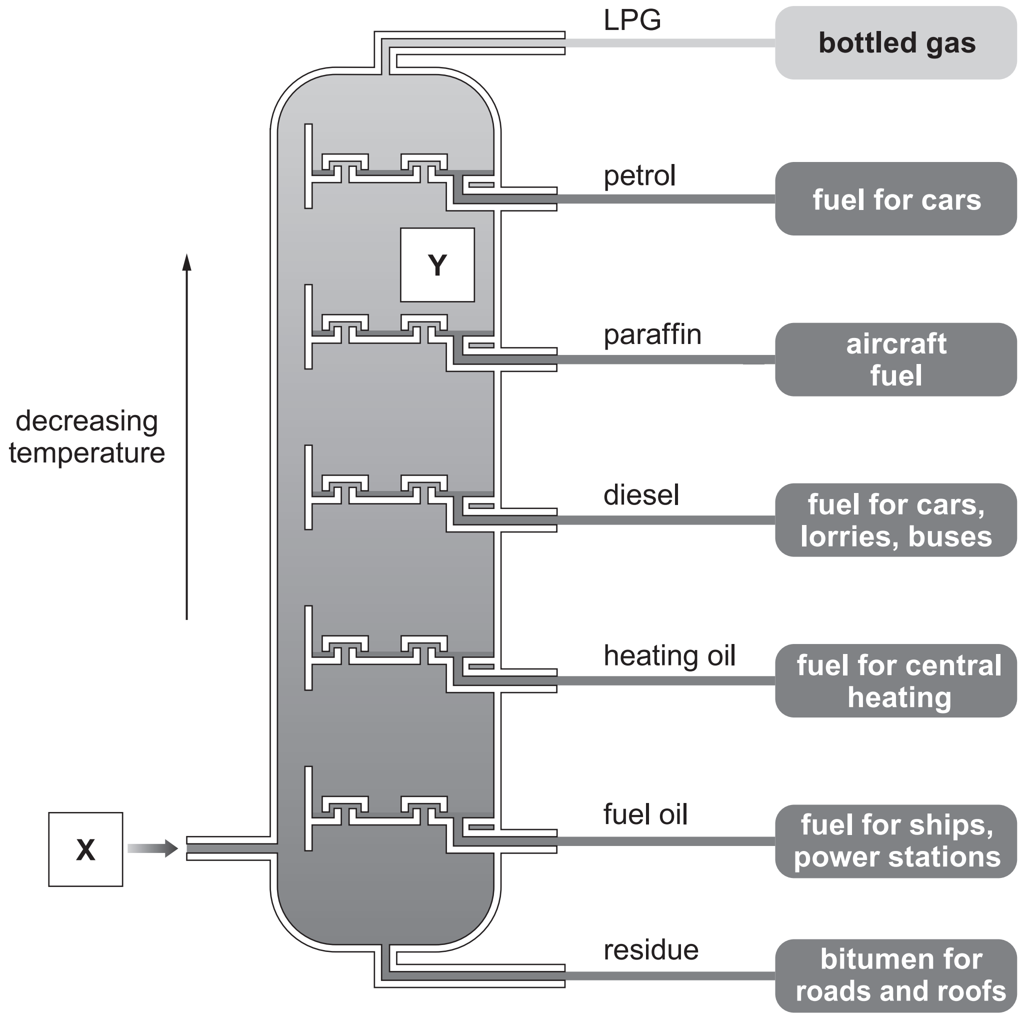
Examiner only

2. (a) Crude oil is separated into mixtures of hydrocarbon compounds in the process of fractional distillation. Many of these fractions are used as fuels.



(i) Name the changes of state happening at **X** and at **Y**. [1]

X ....

Y ....

(ii) Explain why different fractions are formed at different levels. [2]

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(iii) A hydrocarbon fuel was burned and used to heat 100 g of water. The water temperature rose from 18.5 °C to 38.2 °C.

Use the equation below to calculate the amount of energy released by this fuel. Give your answer to **two** significant figures. [3]

energy (J) = mass of water (g) × 4.2 × temperature rise (°C)

Energy = ... J

(b) The products of fractional distillation can undergo a process called cracking to produce smaller, more useful hydrocarbons.

(i) Complete the equation for the cracking of $C_{16}H_{34}$. [1]

$$C_{16}H_{34} \longrightarrow C_8H_{18} + \text{..............................} + C_2H_4$$

(ii) State the **two** conditions used for cracking. [1]

....

(iii) The molecule with the formula C_2H_4 is an unsaturated hydrocarbon.

Give the meaning of the term unsaturated. [1]

....

....

(iv) State why there is a high demand for each of the following products of the cracking reaction. [2]

octane / C_8H_{18} ....

ethene / C_2H_4 ....

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